Lab : Load Balancer Installation And Traffic Simulation (Linux)

Nephtalie Twite Wa Mwamba
202402067
Software Engineering

1 Installation Proof

```
> haproxy -v
HAProxy version 3.2.14-1 2026/03/09 - https://haproxy.org/
Status: long-term supported branch - will stop receiving fixes around Q2 2030.
Known bugs: http://www.haproxy.org/bugs/bugs-3.2.14.html
Running on: Linux 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1
(2026-02-25) x86_64
```

2. Server Startup Evidence

server 1

```
> echo "Server 1 - 202402067" > index.html
> python3 -m http.server 8023
Serving HTTP on 0.0.0.0 port 8023 (http://0.0.0.0:8023/) ...
127.0.0.1 - - [26/Mar/2026 13:20:29] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Mar/2026 13:27:34] "GET / HTTP/1.1" 200 -
```

server 2

```
echo "Server 2 - 202402067" > index.html
python3 -m http.server 8024
Serving HTTP on 0.0.0.0 port 8024 (http://0.0.0.0:8024/) ...
127.0.0.1 - - [26/Mar/2026 13:20:29] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Mar/2026 13:27:34] "GET / HTTP/1.1" 200 -
```

server 3

```
echo "Server 3 - 202402067" > index.html
python3 -m http.server 8025
Serving HTTP on 0.0.0.0 port 8025 (http://0.0.0.0:8025/) ...
127.0.0.1 - - [26/Mar/2026 13:20:29] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Mar/2026 13:27:34] "GET / HTTP/1.1" 200 -
```

3 Unique Server Identify Proof

```
> cat index.html
Server 1 - 202402067

> cat index.html
Server 2 - 202402067

> cat index.html
Server 3 - 202402067
```

4 HAProxy configuration Proof

```
 cat /etc/haproxy/haproxy.cfg | tail  -n 15
        errorfile 500 /etc/haproxy/errors/500.http
        errorfile 502 /etc/haproxy/errors/502.http
        errorfile 503 /etc/haproxy/errors/503.http
        errorfile 504 /etc/haproxy/errors/504.http

frontend http_front
    bind *:8123
    default_backend servers

backend servers
    balance roundrobin
    server s1 127.0.0.1:8023 check
    server s2 127.0.0.1:8024 check
    server s3 127.0.0.1:8025 check
```

5 Load Distribution Evidence

```
for i in {1..10}; do curl http://localhost:8024; done

Server 2 - 202402067
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Server 2 - 202402067
```

6. Load Testing Output

```
> ab -n 100 -c 10 http://127.0.0.1:8123/
This is ApacheBench, Version 2.3 <$Revision: 1923142 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking 127.0.0.1 (be patient).....done


Server Software:        SimpleHTTP/0.6
Server Hostname:        127.0.0.1
Server Port:            8123

Document Path:          /
Document Length:        21 bytes

Concurrency Level:      10
Time taken for tests:   0.043 seconds
Complete requests:      100
Failed requests:        0
Total transferred:      20700 bytes
HTML transferred:       2100 bytes
Requests per second:    2309.63 [#/sec] (mean)
Time per request:       4.330 [ms] (mean)
Time per request:       0.433 [ms] (mean, across all concurrent requests)
Transfer rate:          466.89 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    0   0.1      0       1
Processing:     1    4   1.6      4      10
Waiting:        1    4   1.5      4      10
Total:          1    4   1.6      4      10

Percentage of the requests served within a certain time (ms)
  50%      4
  66%      5
  75%      5
  80%      5
  90%      6
  95%      7
  98%      8
  99%     10
 100%     10 (longest request)
```

7. Faillure Test Evidence

> for i in {1..10}; do curl http://localhost:8123; done
Server 3 - 202402067
Server 1 - 202402067
Server 3 - 202402067
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Server 1 - 202402067

8. Process Verification

> ps aux | grep haproxy
root 10142 0.0 0.4 63156 31948 ? Ss 13:25 0:00 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
haproxy 10144 0.0 1.0 214140 85096 ? Sl 13:25 0:00 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
happy 15543 0.0 0.0 6548 2496 pts/3 S+ 14:03 0:00 grep --color=auto --exclude-dir=.bzr --exclude-dir=CVS --exclude-dir=.git --exclude-dir=.hg --exclude-dir=.svn --exclude-dir=.idea --exclude-dir=.tox --exclude-dir=.venv --exclude-dir=venv haproxy